

SD Core - Introduction and Getting Started

What a multivalued database is, what SD is, the four components, and your first session.

This page orients you to SD Core for Windows: what a multivalued database is, where SD came from, what the pieces are, and how to take your first steps. It is the only page in this set that assumes nothing.

What is a multivalued database?

A multivalued database stores data in records made of **fields**, where each field can hold **more than one value** — and each value can hold **more than one subvalue**. A single field in a customer record can therefore carry every phone number the customer has, without a separate table or a join.

The model was designed by Dick Pick in the 1970s as the Pick Operating System. It has been through PI/open, UniVerse, Unidata, D3, jBASE, QM, and ScarletDME — and SD is one of its direct descendants.

The three delimiters that make it work are **field marks**, **value marks** and **subvalue marks** — control characters that separate the levels inside a single string. A dynamic array in SDBasic is a string that carries these marks, and `extract`, `insert`, `delete` and `replace` work on them directly.

What SD is

SD Core for Windows is a version of SD, with elements found in the main SD version and in ScarletDME. ScarletDME was a fork of the original GPL release of OpenQM 2.6.6.

That lineage matters when you go looking for documentation. Not all the features of the *commercial* OpenQM 2.6.6 were in the GPL release, and no documentation specific to the GPL version was ever released. The OpenQM 2.6.6 documents can be used as a reference, but SD Core has additions, changes and deletions — of features, of structure, of security and of commands. This documentation set covers those changes.

If you have used OpenQM, or SD on Linux, much of SD Core will still be familiar: the same data model, the same query processor, the same BASIC.

SD Core for Windows is Windows only. There are no `#ifdef` branches keeping Linux alive in this source — Linux SD is a separate project and this is not a build of it.

SD Core is free software under the GNU General Public Licence v3. `config_gpl` displays the licence and `config_contrib` the list of contributors. The installer carries compiled binaries; the source is a separate download.

The four components

The command processor (TCL)	reads what you type at the <code>:</code> prompt and dispatches it to a verb, a program, a paragraph or a query
The query processor	runs <code>list</code> , <code>select</code> , <code>count</code> , <code>sort</code> and the rest — the reporting language
SDBasic	the programming language: a compiled BASIC with dynamic arrays, file I/O, and the multivalue string functions
The SDClient API	a C client library (<code>sdc11lib.dll</code>) that lets an external application connect to SD, read and write records, execute commands and call subroutines

Signing in

```
sd
```

You land in **the SD account with your own name**. Nothing asks for a password — Windows has already authenticated you. SD asks Windows who you are.

If `sd` answers *Account ... not in register*, you are in the wrong account or your group membership has not taken effect yet. If it answers *not registered for SD use*, you are not in the `sdusers` group.

You must sign out and back in after being added to `sdusers`. Windows fixes group membership when you sign in. Until you get a new logon token you cannot read the data tree at all, and the symptom looks like a broken install.

SD is already running. It is a Windows service — **String Database (SD)** — and Windows starts it at every boot. You do not type `sd -start`.

Your first file and record

```
create.file customers
ed customers 1001
```

`ed` is the line editor, and it needs nothing installed. In `ed`: `i` to insert, type your lines, a full stop on its own line to stop inserting, then `fi` to file and exit.

A programmer account can also use `edit` (Microsoft Edit, a full-screen editor) or `micro` (a full-screen editor with syntax highlighting). Both need `OS.EXECUTE` permission — see the administrator documentation.

```
list customers
count customers
```

Commands are lower case now. Typing `LIST` still works — SD tries what you typed, then lower case, then upper, and finally with any hyphens changed to dots, so `clear-select` reaches `clear.select` too.

Writing a program

A program lives in a `bp` file — a directory file, which is an ordinary Windows folder with one file per program. You can write it in `ed`, in `edit`, in `micro`, or in any text editor you like (Notepad, VS Code, etc.) — the folder is on disk at:

```
C:\ProgramData\SD\user_accounts\<account>\bp
```

Compile and catalogue it from inside SD:

```
basic bp myprog
catalog bp myprog
```

Then run it by name:

```
myprog
```

Becoming an administrator

```
logto sdsys
```

You will get a UAC consent prompt unless the session is already elevated. That is the gate — there is no SDSYS password, and there is deliberately no second shared secret held by every administrator.

IF YOU ARE OVER ssh, THIS MAY NOT WORK. A UAC prompt has no interactive desktop there. Start an elevated terminal at the machine instead.

What is not in SD Core

The following were in OpenQM, in ScarletDME, or in SD on Linux, and are not in SD Core for Windows:

GONE	WHY
QMNet (remote files)	Removed; the API is the supported way to reach another SD server
Embedded Python	Dropped; the intended use is as a back end data store reached through the API
<code>sdlnxd</code> daemon	Linux-only; the Windows service replaces it
<code>ENCRYPT.FIELD</code> verb	Removed; <code>sdencrypt()</code> and <code>sddecrypt()</code> in SDBasic are the supported route
<code>sed</code> , <code>update.record</code> , <code>modify</code> editors	Gone; use <code>edit</code> , <code>micro</code> or <code>ed</code>
PROC language	Removed; use paragraphs instead
NLS, <code>SET.LANGUAGE</code>	Removed; SD Core is English only
Silent install	Refused deliberately; the installer asks questions that cannot be defaulted
Multi-user Remote Desktop	Not supported; accounts SD creates are denied the console and RDP

Document conventions

bold	a word typed as it stands
<i>italics</i>	something you supply
braces <code>{ }</code>	an optional part
<code>code</code>	a command, a function name, or something you type

